

M-TALHA
SID: 15882
QviZ no-62

5

Qno: 01

Ans: DATA

	x_1	y_1	
Customers	Purchases	Time (hrs)	Category
A	2	1	L
B	3	2	L
C	6	5	H
D	7	6	H
E	5	4	H

New Customer P_1

No. of Purchases = 4 $\rightarrow x_2$

Time spent = 3 $\rightarrow y_2$
 $K=3$

First we find distance

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$A = \sqrt{(4-2)^2 + (3-1)^2}$$

$$= \sqrt{8} \Rightarrow 2.83$$

$$B = \sqrt{(4-3)^2 + (3-2)^2}$$

$$= \sqrt{2} \Rightarrow 1.414$$

$$C = \sqrt{(4-6)^2 + (3-5)^2}$$

$$= \sqrt{8} \Rightarrow 2.83$$

$$D = \sqrt{(4-7)^2 + (3-6)^2}$$

$$= \sqrt{18} \Rightarrow 4.24$$

$$E = \sqrt{(4-5)^2 + (3-4)^2}$$

$$= \sqrt{2} \Rightarrow 1.414$$

Nearest
neighbour.

$K=3$ They have 3 Nearest Neighbours

1.	B	L	→ 1.414	} → 2 Lowest values
2.	E	H	→ 1.414	
3.	A	L	→ 2.83	} → 1 High or sequence

Majority Voting in these neighbours

- $L = 2$
- $H = 1$

Since,

We Using KNN ($K=3$)

Hence

the customer F is classified as:
Low Value [L]. Ans. ✓